Juliette M. Walker - Crown Community Academy

Now Hear This!

Juliette M. Walker
751 E. 84st. Apt#2e
CHICAGO IL 60619
(773) 488-3157

Crown Community Academy
2128 S. St.Louis
CHICAGO IL 60623
(773) 534-1680

Objective(s):

In this project, students will be able to describe how sounds are heard by the human ear by creating a model of the human inner ear. This activity may be utilized by grades 3 - 8.

Materials Needed:

Students may be placed into cooperative learning groups of 3.
1long paper roll (used gift wrap rolls are perfect)
plastic wrap medium sized disc of foil rubber bands
2cardboard paper discs doublestick tape pink tissue paper
flashlights 1 large box scissors cardboard fork shape
(1 set of each item is needed for each group)

Strategy:

1. Students will create an eardrum by stretching a piece of plastic wrap across the end of the paper roll fixing it in place with a rubber band.
2. Students will create a set of ossicles with two cardboard discs and a fork shape of thin cardboard. This will be held together with doublesided tape.
3. Students will then attach a disk of shiny foil to one end of the roll, and will attach the other to the plastic wrap "ear drum" on the tube. This will complete the middle ear.
4. Students will then make an outer ear from a cone of cardboard with a hole at its end and will place it on the inside of the tube. Students may design their cone with pink tissue paper, for a more realistic look.
5. Students will then place the middle of the model of the ear on a box, leaving the ends of the rolls hanging off of the front and back.
6. Students will then shine a light onto the mirror and will ask a partner to to talk into the ear and watch for vibrations.
7. Have the students to watch the window of the ear. Have them describe how it responds to shouting, whispering, whistling etc.

Performance Assessment:

Students will be evaluated based upon their models, and oral explanations as to how their ear works.

Conclusions:

This project works because the sound waves from the students voice makes the plastic wrap vibrate. These vibrations are transmitted into the cardboard "ossicles" and can be seen by watching the foil disc for movements. This is a simple model of how the ear works.

References:

This project was adapted from the **Encyclopedia of Science Projects** written by Pam Robeson, and Mick Seller (Shooting Star Press, 1994) (Aladdin Books, ltd, 1994)

Return to Physics Index